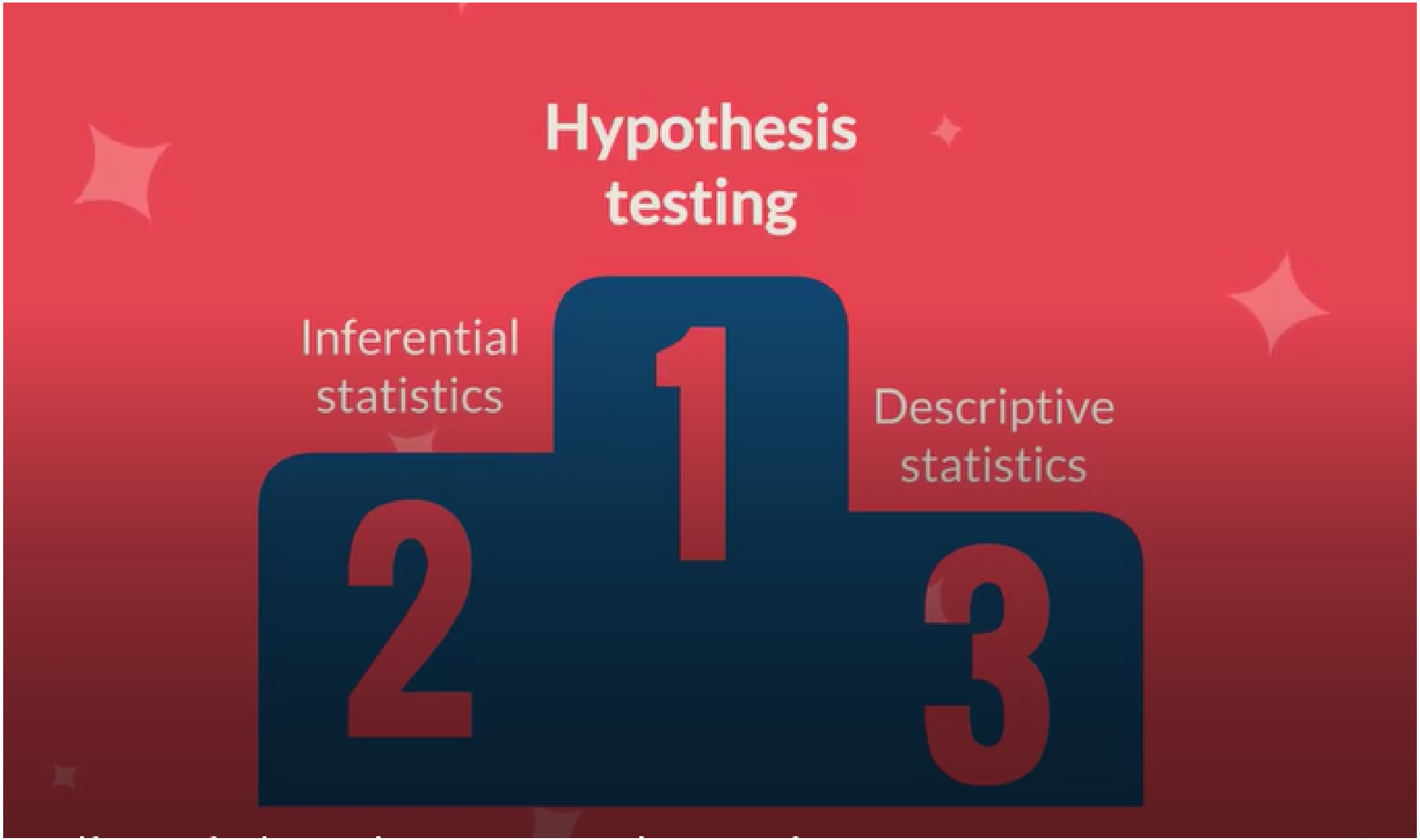

Hypothesis testing
Inferential statistics
1
Descriptive statistics
2
3

Confidence interval



In x% of the cases, the true parameter will fall in the confidence interval

Hypothesis testing

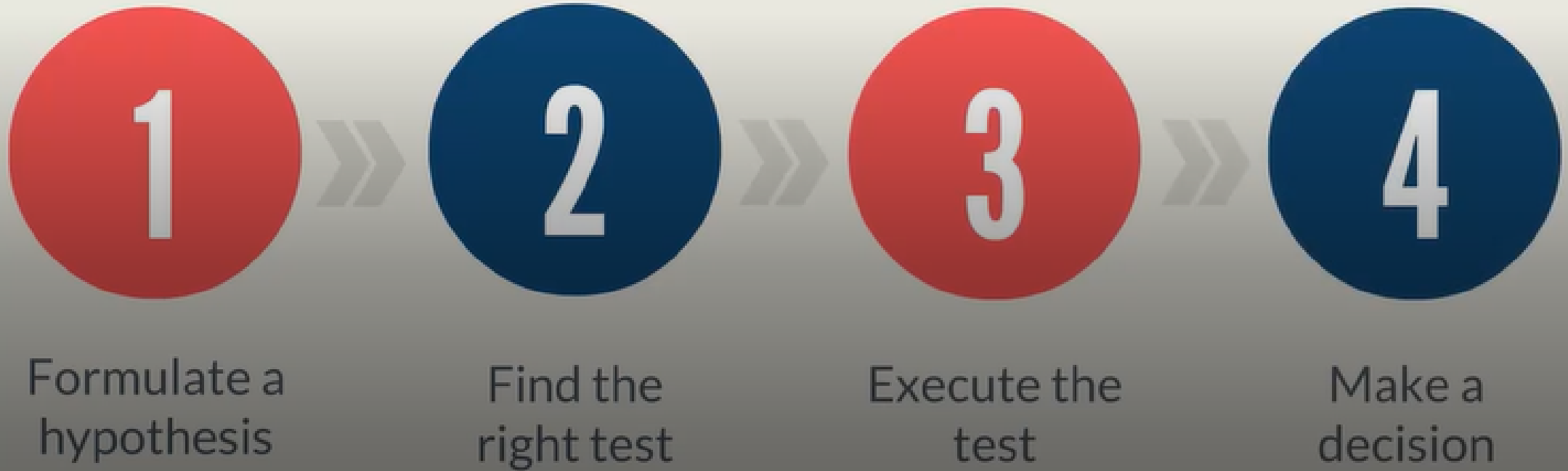


Yes



No

Steps in data-driven decision-making

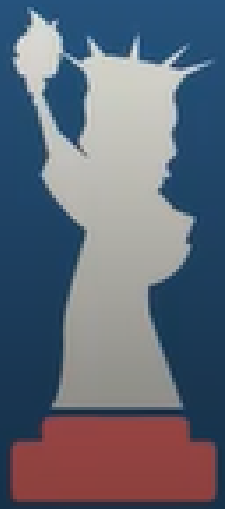


What is a hypothesis?

'A hypothesis is an idea that can be tested'

in  definition

Statement

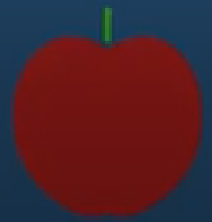




are expensive

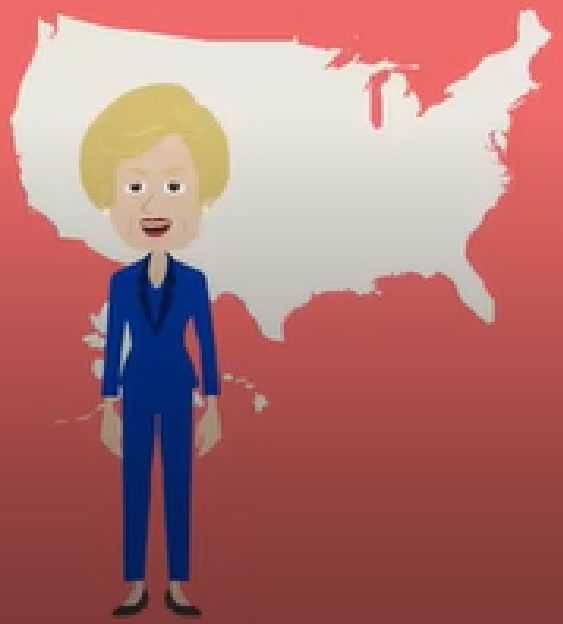
Hypothesis





are more expensive than $1.75

Not a hypothesis



vs.



no data => cannot be tested => not a hypothesis

Example



*Glassdoor is a website where current and former employees rate their employers and the C-level management. All data is self-reported.

Example

Hypotheses	Notation
Null hypothesis	H_0
Alternative hypothesis	H_1 or H_A

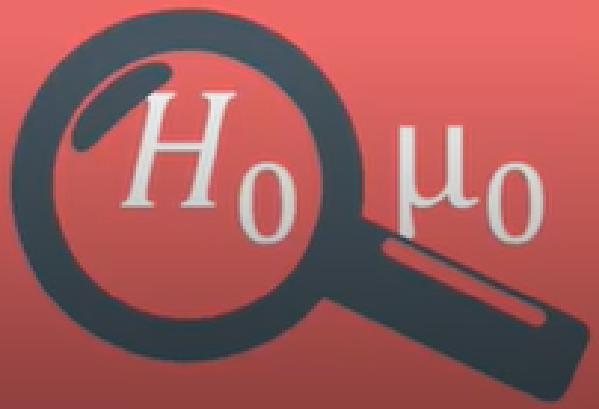


Example

$$H_0: \mu_0 = \$\,113{,}000$$

$$H_1: \mu_0 \neq \$\,113{,}000$$

Example



H_0 $\mu_0 = \$\,113{,}000$

Accept if: μ_0 is close enough to the true mean μ

Reject if: μ_0 is too far from the true mean μ

Example



innocent until proven guilty

H_0 is true until rejected

Example 2.0



Example 2.0

$$H_0: \mu_0 > \$\,125{,}000$$

$$H_1: \mu_0 \leq \$\,125{,}000$$

The researcher is trying to REJECT the null hypothesis.

Example 2.0

status quo $H_0: \mu_0 > \$\,125{,}000$

Change or innnovation $H_1: \mu_0 \leq \$\,125{,}000$



$$H_0: \mu_0 > \$\,125{,}000$$

Change or innnovation



$$H_1: \mu_0 \leq \$\,125{,}000$$

The researcher is trying to REJECT the null hypothesis.